HANJOON SHIM

Ph. D Candidate, Department of Aerospace Engineering, Seoul National University, Seoul, Korea

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SUMMARY

Ph.D candidate has been training in aerospace engineering with strong sociable skills developed from extensive team projects experiences and ability to work leading a team or as part of a team. Special expertise in the following areas:

- *Spacecraft GPS Precise Relative Navigation*
- *Spacecraft Attitude and Orbit Control System*
- *CubeSat (Nanosatellite)*
- *Hardware-In-the-Loop Simulation*

EXPERIENCE

GNSS Laboratory, Seoul National University — Seoul, Korea

Graduate Student Full-time Researcher — *Mar 2017 - Present*

- Contributed to three ***SNUGLITE CubeSat projects***, taking on roles such as subsystem developer, project manager, and technical consultant. Successfully achieved two launches and operational successes (SNUGLITE-I, and SNUGLITE-II), with the current SNUGLITE-III project currently in development.
  - Participated in all stages of system design, developed flight software based on RTOS, and contributed to assembly, integration, testing, and operational phases
  - Developed and implemented a real-time CubeSat attitude determination and control system, established a HILS verification environment for CubeSat platform
  - Developed single frequency GPS RTK-based relative navigation for CubeSat rendezvous
- Developed DGPS relative navigation system as a project manager through one commissioned research.
- Teaching Assistant (M2795.007100: Advanced Aerospace Vehicle Dynamics and Control, M2795.007300: Aerospace Estimation), and provided guidance to four undergraduate students in their research paper endeavors.

Smart Autonomous System Laboratory, Handong Global University — Pohang, Korea

Undergraduate Internship — *Aug 2015 - Jun 2016*

- Teaching assistant (HMM20029: Analog control circuit design, HMM30011: Digital control).
- Presented an academic paper at the 2015 Summer Conference of the Korean Institute of Electrical Engineers.

Republic of Korea Army — Yang-gu, Korea

Sergent, Mandatory Military Service — *Jan 2011 - Oct 2012*

- Developed mental strength through rigorous training and facing North Korea in challenging conditions.
- Gained leadership experience as a squad leader, improved social skills through teamwork with large groups, and honed communication and collaboration skills during mandatory training.

EDUCATION

Seoul National University — Seoul, Korea

Ph.D, Aerospace Engineering | Advisor: Prof. Changdon Kee — *Sep 2019 - present*

- will be submitted in May 2024 (Precise GPS relative navigation for CubeSat rendezvous)

Seoul National University — Seoul, Korea

M.Sc., Mechanical and Aerospace Engineering | Advisor: Prof. Changdon Kee — *Feb 2019*

- HILS verification of low earth orbit cube-satellite attitude determination and control system using Helmholtz cage

Handong Global University — Pohang, Korea

B.Sc., Mechanical and Control Engineering | Advisor: Prof. Wonsang Ra — *Aug 2016*

- Guidance sensors for low-cost drones with autonomous recharging capability

RESEARCH EXPERIENCE

SNUGLITE-III CubeSat Project
GNSS Laboratory, Seoul National University

Technical Advisor, AOCS/FSW/AIT Engineer — *Jul 2022 - Present*

- Finalist in the 2022 Korean CubeSat Contest.
- Developed satellite precise relative navigation for autonomous rendezvous.
  - Designed an RTK relative navigation system (GPS L1 only) with 1cm level accuracy
  - Conducted SILS (Matlab), PILS (Linux-gcc, C), and HILS
- Designed PCB: Solar Panels, Interface Boards, and FlatSat Board.
- Integrated satellite attitude control system.
- Assembly, integration, and test.
- Scheduled for launch in November 2025 (KSLV-II, Nuri).

SNUGLITE-II CubeSat Project *(NORAD 52899)*
GNSS Laboratory, Seoul National University

Team Leader, ADCS/FSW/AIT Engineer — *Jul 2019 - Jun 2022*

- Finalist in the 2019 Korean CubeSat Contest.
- Led the project team and managed project budget, schedule, documents, and contracts.
- Oversaw system design.
- Developed satellite flight software (FreeRTOS and AVR32)
- Designed a satellite momentum wheel-based attitude control system.
  - Conducted SILS (Matlab), PILS (Linux-gcc, C), and HILS
  - Developed a HILS platform for a single-axis attitude control system based on a momentum wheel.
- Designed PCB Interface Boards.
- Assembly, integration, and test.
- Successfully launched the satellite in June 2022 (KSLV-II, Nuri).

KARDSAT CubeSat GPS Relative Navigation Project
GNSS Laboratory, Seoul National University

Project Manager, Navigation Engineer — *Mar 2019 - Dec 2020*

- Conducted commissioned research for the Korean Aerospace Research Institute.
- Developed a real-time single-frequency GPS-based relative navigation system.
  - Designed a Deci-meter level spacecraft DGPS navigation system (GPS L1 only).
  - Conducted SILS (Matlab) and PILS (Linux-gcc, C)
- Developed a C language based PILS environment, establishing a practical verification setup using Matlab, Visual Studio, and OBC communication via UART.

SNUGLITE-I CubeSat Project *(NORAD 43784)*
GNSS Laboratory, Seoul National University

ADCS Engineer — *May 2017 - Dec 2018*

- Finalist in the 2015 Korean CubeSat Contest (participating in the development stage).
- Developed a MEMS-based attitude determination and magnetorquer-based control system.
  - Conducted SILS (Matlab), PILS (Linux-gcc, C), and HILS
- Developed a HILS platform for a single-axis attitude determination and control system, including the design and fabrication of the Helmholtz cage.
- Successfully launched the satellite in December 2018 (Space-X, Falcon-9).

AWARDS

Best Presentation Award – ION GNSS+ 2023 — *Sep 2023*

The Institute of Navigation

Minister Award - 2019 Korea CubeSat Contest — *Nov 2022*

Minister of Science and ICT, Republic of Korea

Director Award - 2022 Korea CubeSat Contest Finalist — *Nov 2022*

Korea Aerospace Research Institute

Best Paper Presentation Award – 2021 KSAS Fall Conference *Apr 2022*

The Korean Society for Aeronautical and Space Sciences (KSAS)

Director Award - 2019 Korea CubeSat Contest Finalist *Feb 2020*

Korea Aerospace Research Institute

Director Award - 2015 Korea CubeSat Contest *Apr 2019*

President of Korea Aerospace Research Institute

Aviation Specialized University Excellent Talent Certificate *Mar 2019*

Minister of Ministry of Land, Infrastructure and Transport, Republic of Korea

Director Award - Capstone Contest *Dec 2015*

School of Mechanical and Control Engineering, Handong Global University

PUBLICATIONS

International Journal Paper (SCIE/SCOPUS)

[1] **H. Shim**, and C. Kee. "Ultra-Low Cost Hardware-In-the-Loop Simulation for CubeSat Platform: Implementation and Verification of Momentum Wheel-Based Attitude Control," (In preparation).

[2] **H. Shim**, and C. Kee. "Highly Efficient Real-Time Kinematic-Based Precise Relative Navigation for Autonomous Rendezvous CubeSat," *Journal of the Institute of Navigation.* (Under review, submitted in Dec 2023).

[3] **H. Shim**, O. Kim, M. Park, M. Choi, and C. Kee. (2023). "Development of Hardware-In-the-Loop Simulation for CubeSat Platform: Focusing on Magnetometer and Magnetorquer," *IEEE Access, Vol. 11, pp. 73164-73179.* https://doi.org/10.1109/ACCESS.2023.3294565

[4] **H. Shim**, O. Kim, S. Yu, C. Kee, D. Cho and H. Kim. (2024). "Precise in-orbit relative navigation technique for rendezvous mission of CubeSats using only GPS receivers," *CEAS Space Journal, Vol. 16. pp. 117-137.* https://doi.org/10.1007/s12567-023-00488-x

[5] M. Choi, O. Kim, H. Kim, **H. Shim**, and C. Kee, (2022). "Probabilistic Determination Algorithm for Detumbling and Angular Rate Convergence using a Three-Axis Magnetometer," *Transactions of the Japan Society for Aeronautical and Space Sciences, Vol. 65, Issue 2, pp. 76-83.* https://doi.org/10.2322/tjsass.65.76

[6] O. Kim, **H. Shim**, S. Yu, Y. Bae, C. Kee, H. Kim, J. Lee, J. Han, S. Han, and Y. Choi. (2020). "In-Orbit Results and Attitude Analysis of the SNUGLITE Cube-Satellite," *Applied Sciences 10, no. 7: 2507*. https://doi.org/10.3390/app10072507s

International Conference Paper

[1] **H. Shim**, B. Kim, Y. Bae, C. Kee, and I. Koo. (2024). "GPS Relative Navigation for Autonomous Rendezvous of SNUGLITE-III CubeSat: Overcoming Challenges in Precision and Efficiency," the 75th International Astronautical Congress (IAC), Milan, Italy, 14-18 October 2024. (Submitted).

[2] **H. Shim**, Y. Bae, C. Kee, and I. Koo. (2024). "Verification and Hardware Implementation of Single-Frequency RTK-Based Precise Relative Navigation for SNUGLITE-III CubeSat," *Proceedings of the 2024 International Technical Meeting of The Institute of Navigation (ION ITM 2024), Long Beach, California, Jan 2024, pp. 787-798*. https://doi.org/10.33012/2024.19486

[3] **H. Shim**, Y. Bae, J. Hwang, C. Kee and I. Koo. (2023). "Single Frequency RTK Relative Navigation for Autonomous Formation Flying Mission of SNUGLITE-III CubeSat," *Proceedings of the 36th International Technical Meeting of the Satellite Division of The Institute of Navigation (ION GNSS+ 2023), Denver, Colorado, Sep 2023, pp. 1474-1486.* https://doi.org/10.33012/2023.19371 BEST PRESENTATION

[4] J. Park, **H. Shim**, Y. Bae, C. Kee, and S. Yu. (2023). "Verification of Space GPS Receiver Navigation Performance Based on Full-Orbit Navigation Solutions of CubeSat," *Proceedings of the 2023 International Technical Meeting of The Institute of Navigation (ION ITM 2023), Long Beach, California, Jan 2023, pp. 252-261.* https://doi.org/10.33012/2023.18600

[5] S. Yu, O. Kim, **H. Shim**, Y. Bae, C. Kee, H. Kim, J. Lee, J. Han, S. Han, Y. Choi, and W. Choi. (2020). "Verification of Space GPS Receiver Navigation Performance Using a Cube Satellite," *Proceedings of the 2020 International Technical Meeting of The Institute of Navigation (ION ITM 2020), San Diego, California, Jan 2020, pp. 736-744.* https://doi.org/10.33012/2020.17174

[6] M. Choi, S. Yu, O. Kim, H. No, **H. Shim**, and C. Kee. (2018). "Development and Verification of Attitude Determination and Control Algorithm for SNUGLITE Cube Satellite," *Proceedings of the 2018 International Technical Meeting of The Institute of Navigation (ION ITM 2018), Reston, Virginia, Jan 2018, pp. 56-71.* https://doi.org/10.33012/2018.15534

Domestic Journal Paper

[1] H. Shim, Y. Bae, J. Park, H. Jeong, J. Lee, O. Kim, C. Kee, K. Kwon, C. Koo, and I. Koo. (2023). "Operation Results of SNUGLITE-II CubeSat: Convective 3D Observation and Precise Attitude Determination Using GPS Carrier Measurements," *Journal of the*

Korean Society for Aeronautical & Space Sciences, Vol. 51, Issue 10, pp. 721-731. https://doi.org/10.5139/JKSAS.2023.51.10.721 SELECTED BEST PAPER

[2] **H. Shim**, Y. Bae, and C. Kee. 2023. "Real-time Software Verification Technique of Attitude Determination and Control System for CubeSat," *Journal of the Korean Society for Aeronautical & Space Sciences, Vol. 51, Issue 3, pp. 207-216.* https://doi.org/10.5139/JKSAS.2023.51.3.207

[3] **H. Shim**, O. Kim, S. Yu, C. Kee, D. Cho and H. Kim. (2020). "Single Frequency GPS Relative Navigation for Autonomous Rendezvous and Docking Mission of Low-Earth Orbit Cube-Satellites," *Journal of Positioning, Navigation, and Timing 9, no.4, pp. 357-366.* https://doi.org/10.11003/JPNT.2020.9.4.357

[4] M. Choi, J. Jang, S. Yu, O. Kim, **H. Shim**, and C. Kee. (2017). "Single-Axis Hardware in the Loop Experiment Verification of ADCS for Low Earth Orbit Cube-Satellite," *Journal of Positioning, Navigation, and Timing 6, no. 4 pp. 195–203.* https://doi.org/10.11003/JPNT.2017.6.4.195

Domestic Conference Paper

~34 Papers have been presented (15 as the first author presentations, 19 as co-authors)

- **H. Shim**, and C. Kee. (2021). "HILS Verification of SNUGLITE-II Cube Satellite Attitude Control System Equipped with Reaction Wheel and Magnetorquer," *Proceedings of the 2021 Korean Society for Aeronautical & Space Sciences Fall Conference, Jeju, Korea, Nov 2021, pp. 321-322*. BEST PAPER PRESENTATION

OTHERS

Programming	C/C++, Matlab, FreeRTOS
Technical	Embedded (AVR), CAD (CATIA), PCB (OrCAD/Allegro), Photo editing (Photoshop, Lightroom), Third-class armature radio operator (Telephone)
Language	Korean (Native), English (Upper Intermediate), Japanese (elementary)
Interests	Shutter Stock Photographer, Aggressive Skier

REFERENCES

Professor. Changdon Kee (Master's, Ph.D. advisor)
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Room#1321, Build#301, 1, Gwanak-ro, Gwanak-gu, Seoul, Korea, 08826
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Professor. O-Jong Kim (SNUGLITE-I Project team leader)
DEPARTMENT OF AEROSPACE ENGINEERING, SEJONG UNIVERSITY
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